

Dario Gherzi

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Education

Postdoctoral Training, Princeton University, 2010-2014

Ph.D. Computational Biology, New York University/Mount Sinai, 2010

M.D. University of Genoa, Italy, 2004, 110/110 *Summa Cum Laude*

Honors and Awards

College of IS&T Research Award (2025)

National Strategic Research Institute Fellow, 2021-present

American-Italian Cancer Foundation Postdoctoral Fellowship, 2011-2013

NBCR Scholar Fellowship, 2007, University of California, San Diego

Medal for academic achievements, 2004, University of Genoa

Experience

Associate Professor of Biomedical Informatics	2020-present
School of Interdisciplinary Informatics	University of Nebraska at Omaha

Assistant Professor of Biomedical Informatics	2014-2020
School of Interdisciplinary Informatics	University of Nebraska at Omaha
Development of computational methods to study immunoinformatics and cancer genomics datasets.	
Structural bioinformatics of protein binding.	

Postdoctoral Fellow	2010-2014
Mona Singh	Princeton University
Development of computational methods for the analysis of protein interactions and their alterations in pathology.	

PhD Student	2006-2010
Roberto Sanchez	Mount Sinai School of Medicine
Development of computational methods to detect and characterize protein binding sites	

Research Assistant	2004-2005
Franco Celada	NYU Medical Center
Mathematical modeling of the immune system with a multiagent-based simulator. Development of routines to model crossreactivity in T cell-mediated immune response to viral infections.	

Teaching Experience

Courses taught at the University of Nebraska at Omaha:

- ACMP-2000 “Data Analysis and Machine Learning” (newly developed course)
- BIOI-1000 “Introduction to Bioinformatics”
- BIOI-2000 “Foundations of Bioinformatics”
- BIOI-3500 “Advanced Bioinformatics Programming”
- HONR-3040 “Exploring Complex Systems with Computer Programs” (newly developed course)
- BMI-8100 “Introduction to Biomedical Informatics”
- BMI-8300 “Public Health Genomics” (newly developed course)
- BMI-8400 “Linear Algebra for Advanced Computing and AI” (newly developed course)
- BMI-8850 “Biomedicine for the Nonmedical Professional” (newly developed course)

First-author journal articles

Gherzi D, Matthos T, “Identifying Molecular Fragments That Drive 7-Dehydrocholesterol Elevation”, *ACS Pharmacol. Transl. Sci.*, 2022, 5, 1, 3–7

Gherzi D, Parakh A, Mezei M, “Comparison of a quantum random number generator with pseudo-random number generators for their use in molecular Monte Carlo simulations”, *Journal of Computational Chemistry*, 2017, 10.1002/jcc.25065

Gherzi D, Singh M, “molBLOCKS: a suite to break down small molecules into fragments and analyze their distribution”, *Bioinformatics* 2014, 10.1093/bioinformatics/btu173

Gherzi D, Singh M, “Interaction-based discovery of functionally important genes in cancers”, *Nucleic Acids Research* 2013, 10.1093/nar/gkt1305

Gherzi D, Singh M, “Disentangling the complex relationship between function, network features and disease: an application to cancer and Mendelian disorders”, *BMC Systems Biology* 2013, 7:5

Gherzi D, Sanchez R, “Automated Identification of Binding Sites for Phosphorylated Ligands in Protein Structures”, *Proteins* 2012, 80:2347-58

Gherzi D, Sanchez R, “Beyond structural genomics: computational approaches for the identification of ligand binding sites in protein structures”, *Journal of Structural and Functional Genomics* 2011, 12(2):109

Gherzi D, Sanchez R, “EasyMIFs and SiteHound: tools for the identification of ligand-binding sites in protein structures”, *Bioinformatics* 2009, 25(23):3185-6

Gherzi D, Sanchez R, “Improving accuracy and efficiency of blind protein-ligand docking by focusing on predicted binding sites”, *Proteins* 2009, 4(2):417-24

Corresponding/Co-corresponding author journal articles

Glynn E, Gherzi D, Singh M (co-corresponding author), “Towards Equitable MHC Binding Predictions: Computational Strategies to Assess and Reduce Data Bias”, *PNAS*, 2025, 122 (8) e2405106122

Klingenberg AS, Gherzi D, “VIPER: Virus Inhibition via Peptide Engineering and Receptor-Mimicry”, *Journal of Computational Biology*, 2025, <https://doi.org/10.1089/cmb.2024.0866>

Seamann A, Bennett-Boehm M, Ehrlich R, Gil A, Selin LK, Gherzi D, “TRain: T-cell receptor automated immunoinformatics”, *BMC Bioinformatics* 2025, 26:76

Ehrlich R, Glynn E, Singh M, Gherzi D, “Computational Methods for Predicting Key Interactions in T Cell-Mediated Adaptive Immunity”, *Annual Review of Biomedical Data Science*, 2024, 7

Thapa I, Gherzi D, “Modeling preferential attraction to infected hosts in vector-borne diseases”, *Frontiers in Public Health* 11, 1276029

- Seamann A, Renzi F, Ganguly K, Pandey K, Byrareddy S, Batra S, Kumar S, [Gheri D](#), “Engineering an ACE2-derived fragment as a decoy for novel SARS-Cov-2 virus”, *ACS Pharmacology & Translational Science*, 2023, 6, 6, 857–867
- Chirravuri-Venkata R, Dam V, Nimmakayala R, Alsafwani Z, Bhyravbhatla N, Lakshmanan I, Ponnusamy M, Kumar S, Jain M, [Gheri D](#), Batra S, “MUC16 and TP53 family co-regulate tumor-stromal heterogeneity in pancreatic adenocarcinoma”, *Frontiers in Oncology*, 2023, 13:1073820.
- Gasper W, Rossi F, Ligorio M, [Gheri D](#), “Variant calling enhances the identification of cancer cells in single-cell RNA sequencing data”, *PLOS Computational Biology* 2022, e1010576.
- Ehrlich R, Kamga L, Gil A, Luzuriaga K, Selin LK, [Gheri D](#), “SwarmTCR: a computational approach to predict the specificity of T Cell Receptors”, *BMC Bioinformatics* 2021, 22, 422
- Chaudhary S, Dam V, Ganguly K, Sharma S, Atri P, Chirravuri-Venkata R, Cox JL, Sayed Z, Jones DT, Ganti AK, [Gheri D](#), Macha MA, Batra SK, “Differential mutation spectrum and immune landscape in African Americans versus Whites: A possible determinant to health disparity in head and neck cancer”, *Cancer letters* 2020, 492:44-53
- West S, Ali H, [Gheri D](#), “NEEP: null empirically estimated p-values for high-throughput molecular survival analysis”, *Journal of Open Source Software* 2020, 5(52):2044
- West S, Kumar S, Batra SK, Ali H, [Gheri D](#), “Uncovering and characterizing splice variants associated with survival in lung cancer patients”, *PLOS Computational Biology* 2019, e1007469.
- Carmicheal J, Atri P, Sharma S, Kumar S, Chirravuri Venkata R, Kulkarni P, Salgia R, [Gheri D](#), Kaur S, Batra SK, “Presence and structure?activity relationship of intrinsically disordered regions across mucins”, *FASEB Journal* 2019, 34 (2), 1939-1957
- Hale ML, Thapa I, [Gheri D](#), “FunSet: an open-source software and web server for performing and displaying Gene Ontology enrichment analysis”, *BMC Bioinformatics* 2019, 20(1): 359
- Scott McGrath and [Dario Gheri](#), “Building towards Precision Medicine: empowering medical professionals for the next revolution”, *BMC Medical Genomics* 2016, 9:23
- Sunandini Sharma, Kritika Karri, Ishwor Thapa, Dhundy Bastola, and [Dario Gheri](#), “Identifying enriched drug fragments as possible candidates for metabolic engineering”, *BMC Medical Genomics* 2016, 9(Suppl 2):46
- Pritykin Y, [Gheri D](#), Singh M, “Genome-wide Detection and Analysis of Multifunctional Genes”, *PLOS Computational Biology* (2015), 10.1371/journal.pcbi.1004467

Other journal publications

- Newcomer K, Bifulco A, Reeves S, Pontecorvi G, Asawa S, Sznurkowska MK, Nuesch M, Ozimski LL, Bolm L, Sieren MM, Muller L, Chi Z, Gasper W, Wachsmann MB, Moore W, Zheng X, Fokkema A, Ring A, Nguyen-Strauli BD, Budinjas S, ten Winkel M, Farber B, Zaki A, Fatimah N, Crecca E, Karalis JD, Chandra R, Prince H, Goff KL, Choi S, Abousaab C, Zhang S, Cole S, Naumann K, Keck T, Polanco PM, Gvozdenovic A, Krol I, Castro-Giner F, Arteaga C, Zeh HJ, Kloeckner R, [Gheri D](#), Aceto N, Ligorio M, “Macrovascular tumor infiltration and circulating tumor cell cluster dynamics in patients with cancer approaching end of life”, *Nature Medicine* 2025, in press
- Kisling SG, Atri P, Shah A, Cox JL, Sharma S, Smith LD, [Gheri D](#), Batra SK, “A Novel HOXA10-Associated 5-Gene-Based Prognostic Signature for Stratification of Short-term Survivors of Pancreatic Ductal Adenocarcinoma”, *Clinical Cancer Research* 2023, 29 (18), 3759-3770
- Ganguly K, Cox JL, [Gheri D](#), Grandgenett PM, Hollingsworth MA, Jain M, Kumar S, Batra SK, “Mucin 5AC-mediated CD44/ITGB1 clustering mobilizes adipose-derived mesenchymal stem cells to modulate pancreatic cancer stromal heterogeneity”, *Gastroenterology* 2022, 162 (7), 2032-2046
- Mischke J, Klein S, Seamann A, Prinz I, Selin L, [Gheri D](#), Cornberg M, Kraft A, “Cross-Reactive T Cell Response Exists in Chronic Lymphocytic Choriomeningitis Virus Infection upon Pichinde Virus Challenge”, *Viruses* 2022, 14(10): 2293

- Clarke MA, Gherzi D, “Electronic health record (EHR) simulation into biomedical informatics course improves students’ understanding of the impact of EHR documentation burden and usability”, *Health and Technology* 2022, 12(2): 465-472
- Clark F, Gil A, Thapa I, Aslan N, Gherzi D, Selin LK, “Cross-reactivity influences changes in human influenza A virus and Epstein Barr virus specific CD8 memory T cell receptor alpha and beta repertoires between young and old”, *Frontiers in Immunology* 2022, 13: 1011935
- Thompson CM, Cannon A, West S, Gherzi D, Atri P, Bhatia R, Smith L, Rachagani S, Wichman C, Kumar S, Batra SK, “Mucin Expression and Splicing Determine Novel Subtypes and Patient Mortality in Pancreatic Ductal Adenocarcinoma”, *Clinical Cancer Research*, 27 (24): 6787–6799
- Cannon A, Thompson CM, Carlo Maurer H, Atri P, Bhatia R, West S, Gherzi D, Olive KP, Kumar S, Batra SK, “CXCR3 and Cognate Ligands are Associated with Immune Cell Alteration and Aggressiveness of Pancreatic Ductal Adenocarcinoma”, *Clinical Cancer Research* 2020, 26 (22), 6051-6063
- Klein S, Gherzi D, Manns MP, Prinz I, Cornberg M, Kraft ARM “PD-L1 Checkpoint Inhibition Narrows the Antigen-Specific T Cell Receptor Repertoire in Chronic Lymphocytic Choriomeningitis Virus Infection”, *Journal of Virology* 2020, 94(18)
- Gil A, Kamga L, Chirravuri-Venkata R, Aslan N, Clark F, Gherzi D, Luzuriaga K, Selin LK, “Epstein-Barr Virus Epitope?Major Histocompatibility Complex Interaction Combined with Convergent Recombination Drives Selection of Diverse T Cell Receptor α and β Repertoires”, *mBIO* 2020, 11(2)
- Gautam SK, Kumar S, Dam V, Gherzi D, Jain M, Batra SK, “MUCIN-4 (MUC4) is a novel tumor antigen in pancreatic cancer immunotherapy”, *Seminars in Immunology* 2020, 47:101391
- Kamga L, Gil A, Song A, Brody, Gherzi D, Aslan N, Stern L, Selin LK, Luzuriaga K, “CDR3 α drives selection of the immunodominant Epstein Barr virus (EBV) BRLF1-specific CD8 T cell receptor repertoire in primary infection”, *PLOS Pathogens* 2019, e1008122
- Castiglione F, Gherzi D, Celada F, “Computer Modeling of Clonal Dominance: Memory-Anti-Naïve and Its Curbing by Attrition”, *Frontiers in Immunology* 2019, 10:1513
- Munro D, Gherzi D, Singh M, “Two critical positions in zinc finger domains are heavily mutated in three human cancer types”, *PLOS Computational Biology* 2018, doi.org/10.1371/journal.pcbi.1006290
- Aslan A, Watkin L, Gil A, Mishra R, Clark F, Welsh R, Gherzi D, Luzuriaga K, Selin LK, “Severity of Acute Infectious Mononucleosis Correlates with Cross-Reactive Influenza CD8 T-Cell Receptor Repertoires”, *mBIO*, 2017, 10.1128/mBio.01841-17.
- Watkin L, Mishra R, Gil A, Aslan N, Gherzi D, Luzuriaga K, Selin LK, “Unique influenza A cross-reactive memory CD8 T-cell receptor repertoire has a potential to protect against EBV seroconversion”, *Journal of Allergy and Clinical Immunology*, 2017, S0091.
- Song I, Gil A, Mishra R, Gherzi D, Selin LK, Stern L, “Broad TCR repertoire and diverse structural solutions for recognition of an immunodominant CD8⁺ T cell epitope”, *Nature Structural & Molecular Biology* 2017, 24:395
- Chakravarty S, Gherzi D, Sanchez R, “Systematic Assessment of Accuracy of Comparative Model of Proteins Belonging to Different Structural Fold Classes”, *J Mol Mod* 2011, 17(11):2831
- Calcagno C, Puzone R, Pearson YE, Cheng Y, Gherzi D, Selin LK, Welsh RM, Celada F “Computer simulations of heterologous immunity: highlights of an interdisciplinary cooperation”, *Autoimmunity* 2011, 44(4):304
- Zhang Q, Chakravarty S, Gherzi D, Plotnikov AN, Sanchez R, Zhou MM, “Biochemical Profiling of Histone Binding Selectivity of the Yeast Bromodomain Family”, *PLoS One* 2010, 5(1):e8903
- Hernandez M, Gherzi D, Sanchez R, “SITEHOUND-web: a server for ligand binding site identification in protein structures”, *Nucleic Acids Research* 2009, 37(Web Server issue): W413-W416

Cheng Y, Gherzi D, Calcagno C, Selin LK, Puzone R, Celada F, “A discrete computer model of the immune system reveals competitive interactions between the humoral and cellular branch and between cross-reacting memory and naive responses”, *Vaccine* 2009, 27(6):833-45

Bahl K, Kim SK, Calcagno C, Gherzi D, Puzone R, Celada F, Selin LK, Welsh RM “IFN-induced attrition of CD8 T cells in the presence or absence of cognate antigen during the early stages of viral infections”, *J Immunol* 2006, 176(7):4284-95

Cornberg M, Chen AT, Wilkinson LA, Brehm MA, Kim SK, Calcagno C, Gherzi D, Puzone R, Celada F, Welsh RM, Selin LK “Narrowed TCR repertoire and viral escape as a consequence of heterologous immunity”, *J Clin Invest* 2006, 116(5):1443-56

Selin LK, Cornberg M, Brehm MA, Kim SK, Calcagno C, Gherzi D, Puzone R, Celada F, Welsh RM “CD8 memory T cells: cross-reactivity and heterologous immunity”, *Semin Immunol* 2004 16(5):335-47

Conference proceedings

Gaspar W, MA J, Gherzi D, Gnimpieba EZ, , Gadhamshetty, Chundi P “Automatic Extension of Medical Subject Headings (MeSH) Thesaurus to Emerging Research”, *2021 IEEE International Conference on Bioinformatics and Biomedicine*

Chetti P*, Ali H[‡], Gandhi R[‡], Ricks B[‡], Gherzi D, Najjar L[‡], “A new approach for analyzing safety and performance factors in civil infrastructures using correlation networks and population analysis”, 2021, *Proceedings of the 13th IWSHM*.

Gaspar W, Chundi P, Gherzi D, “MeSH Indexing Using the Biomedical Citation Network”, *11th ACM Conference on Bioinformatics*, 2020

Li Q*, Gherzi D, Thapa I, Zhang L, Ali H[‡], Cooper K[‡], “Identifying structural changes in correlation networks models of cancer gene expression by stage”, *IEEE BIBM* 2019.

Castiglione F, Gherzi D, Celada F, “In-Silico analysis of the “memory anti-naïve” effect in anti-viral cross-reactive responses”, *IEEE BIBM* 2017

Ehrlich R, Gherzi D, “Analyzing T cell receptor alpha/beta usage in binding to the pMHC”, *IEEE BIBM* 2017

Gherzi D, Sanchez R, “Recovering bound forms of protein structures using the Elastic Network Model and Molecular Interaction Fields”, *7th ACM Conference on Bioinformatics* 2016

Cooper KM[‡], Bastola DR[‡], Gandhi R[‡], Gherzi D, Hinrichs S, Morien M, Fruhling A[‡], “Forecasting the spread of mosquito-borne disease using publicly accessible data: a case study in Chikungunya”, *AMIA 2016 Annual Symposium*.

Balasubramanya A*, Thapa I, Bastola D[‡], Gherzi D, “A novel approach to identify shared fragments in drugs and natural products”, *IEEE BIBM* 2015.

Book chapters

Chirravuri Venkata R, Gherzi D, “Biological pathway data formats and standards”, book chapter, *Elsevier Encyclopedia of Bioinformatics*, 2017

Chirravuri Venkata R, Gherzi D, “Biological pathway analysis”, book chapter, *Elsevier Encyclopedia of Bioinformatics*, 2017

Sharma A, Ali H, Gherzi D, “Cluster analysis of biological networks”, book chapter, *Elsevier Encyclopedia of Bioinformatics*, 2017

Citation Count

(Data obtained from Google Scholar in September of 2025)

Total number of citations: 2,191

h-index: 23

i10-index: 34

Grants

1. NIH NE-INBRE, “Nebraska Research Network in Functional Genomics”, \$180,185 to UNO, with University of Nebraska Medical Center
2. NIH R37, (UNO PI), “Targeting tumor architecture as a novel therapeutic strategy for pancreatic cancer”, \$513,993 to UNO, with University of Texas Southwestern
3. NIH R01, (UNO PI), “Altered T Cell Responses in Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS)”, \$351,885 to UNO, with University of Massachusetts Chan Medical School
4. NIH R01, (UNO PI), “Computational methods to decipher somatic alterations in cancer genomes”, \$116,630 to UNO, with Princeton University
5. NIH R25, (Co-Investigator), “E-PACERR: Enhancing Professionalism, Advocacy, and Capacity For Excellence In Responsible Bioinformatics Research \$262,068
6. NSF EPScoR Grant, (PI), “Acquisition of a Dedicated GPU Rack Server for Bioinformatics”, \$94,520
7. NU Collaboration Initiative, (UNO PI), “Splice Variants as Biomarkers and Therapeutic Targets”, \$71,125 to UNO, with University of Nebraska Medical Center
8. UCRCA Grant, (PI), “Subfields and Areas of Interest in Bioinformatics”, \$5,000

Invited Talks

“*Form follows function*: a novel paradigm of tumor progression in pancreatic adenocarcinoma”, School of Algorithmic Cancer Biology, National Cancer Institute, NIH Campus, Bethesda, MD, 2025.

“Computational approaches for analyzing -omics data in cancer: from bulk sequencing to spatial transcriptomics”, Indiana University, Indianapolis, IN, 2023.

“Computational approaches for analyzing -omics data in cancer”, Simmons Cancer Center, UT Southwestern, Dallas, TX, 2023.

“Computational approaches for analyzing -omics data in cancer”, St. Louis University, St. Louis, MO, 2023.

“Computational approaches to characterize antiviral immune responses”, Big Data Statistical Methodology Workshop, Omaha, NE, 2018.

“Computational Approaches in Cancer Genomics and Heterologous Immunity”, Dept. of Biochemistry, University of Nebraska Medical Center, 2017.

“From mutations to drugs: a computational perspective”, South Dakota State University, Brookings, SD, 2016.

”Omics in biomedicine: bridging the gap between statistical analysis and mechanistic interpretation”, Big Data for Health and Medicine Workshop, Omaha, NE, 2016.

“Making life computable: the challenges of data-driven biology”, ETS, Princeton, NJ, 2013.

“Protein interactions: from molecules to networks”, Nanyang Technological University, Singapore, 2012.

“Increasing accuracy and speed of protein-ligand docking approaches”, University of Sheffield, U.K., 2009.

“Computational approaches to detect and exploit protein binding sites”, Cold Spring Harbor, NY, 2008.

Service

Associate Editor, *BMC Bioinformatics*, 2020 – present

Associate Editor, *Journal of Computational Biology*, 2020 – present

Co-director of the IGPBS Doctoral Program (Bioinformatics track), UNMC, 2019 – 2025

Chair of the Biomedical Informatics Graduate Program Committee at UNO, 2018 – 2023

Chair of the Graduate Program Committee in the School of Interdisciplinary Informatics at UNO, 2023 – present

Chair of the Bioinformatics Undergraduate Program Committee at UNO, 2016 – 2018

Member of the Doctoral Program Committee in Biomedical Informatics at UNO, 2015 – present

NIH grant review panelist for the GCAT (Genomics, Computational Biology and Technology) study session, 2018 – present

Member of the University Committee for the Advancement of Teaching at UNO, 2018 – present

Member of the University Committee on Research and Creative Activity at UNO, 2015 – 2018

Referee for *PLoS Computational Biology*, *Nucleic Acids Research*, *Bioinformatics*, *IEEE Transactions on Computational Biology and Bioinformatics*, *BioMed International*, *Journal of Chemoinformatics*.

Software Development

SwarmTCR – tool for predicting the specificity of T Cell Receptors

(<https://github.com/thecodingdoc/swarmTCR>)

NEEP – tool for performing high-throughput survival analysis (<https://github.com/thecodingdoc/neep>)

FunSet – for performing Gene Ontology enrichment analysis calculations and visualization (in collaboration with Dr. Matt Hale). (<https://github.com/thecodingdoc/GOUtil>, <http://funset.uno>)

molBLOCKS – for breaking down small molecules into fragments and analyzing their composition (<http://compbio.cs.princeton.edu/molblocks>)

CanBind – for exploring protein binding sites mutated in human cancers (<http://canbind.princeton.edu>)

fcsampling – for carrying out functionally-constrained sampling (<http://compbio.cs.princeton.edu/fcsampling>)

EasyMIFs and **SiteHound** – for computing Interaction Energy Maps and performing binding site identification in protein structures (<http://sitehound.sanchezlab.org>)

Skills

Programming languages: C/C++, Python, R, Lisp, Octave/Matlab, Mathematica, NetLogo

Operating systems: Linux, Mac OS X, Windows

Database design and implementation (MySQL)

Software: MODELLER (comparative modeling), AutoDock (molecular docking)

References

Prof. Mona Singh, (mona@cs.princeton.edu), Dept. of Computer Science and Lewis-Sigler Institute for Comparative Genomics, Princeton University, Princeton, NJ, Tel. +1 609-258-7059

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Prof. Robert Hammer (robert.hammer@utsouthwestern.edu), Dept. of Biochemistry, UT Southwestern, Dallas, TX, Tel. +1 214-645-6292